LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuses, current limiters, limiting resistors, circuit breakers, reverse current cut-out relays, reverse current circuit breakers Safeties to be taken installing fuses & circuit breakers. Practical on fuse & circuit breaker.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuses, current limiters, limiting resistors, circuit breakers, reverse current cut-out relays, reverse current circuit breakers Safeties to be taken installing fuses & circuit breakers. Practical on fuse & circuit breaker.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Installation of fuses and circuit breakers; Safety precautions regarding live circuits; Instructor supervision during practical handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a circuit protection device?
(b) Name two types of overcurrent protection devices used in aircraft.
(c) Why is it critical to match the fuse rating to the circuit load?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fuses and current limiters • Limiting resistors function • Circuit breaker operation • Reverse current cut-out relays • Reverse current circuit breakers	L & Ds	PPT, IB	15 min
(b)	• Safeties to be taken • Installing fuses procedures • Installing circuit breakers • Troubleshooting protection devices • Maintenance and inspection	L & Ds	PPT, PDF	20 min
(c)	• Practical on fuse installation • Practical on circuit breaker installation • Verification of correct installation • Safety checks post-installation • Instructor feedback and correction	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a fuse and a circuit breaker in terms of reusability.
(b) Describe the safety precautions required when installing a reverse current circuit breaker.
(c) How does a limiting resistor protect a circuit from overcurrent?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between DC and AC overvoltage protection systems to be discussed in LP 14.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Over voltage protection DC generates systems, over voltage protection AC generating systems, under voltage protection, under and over excitation protection, under and over frequency protection, differential current protection.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min